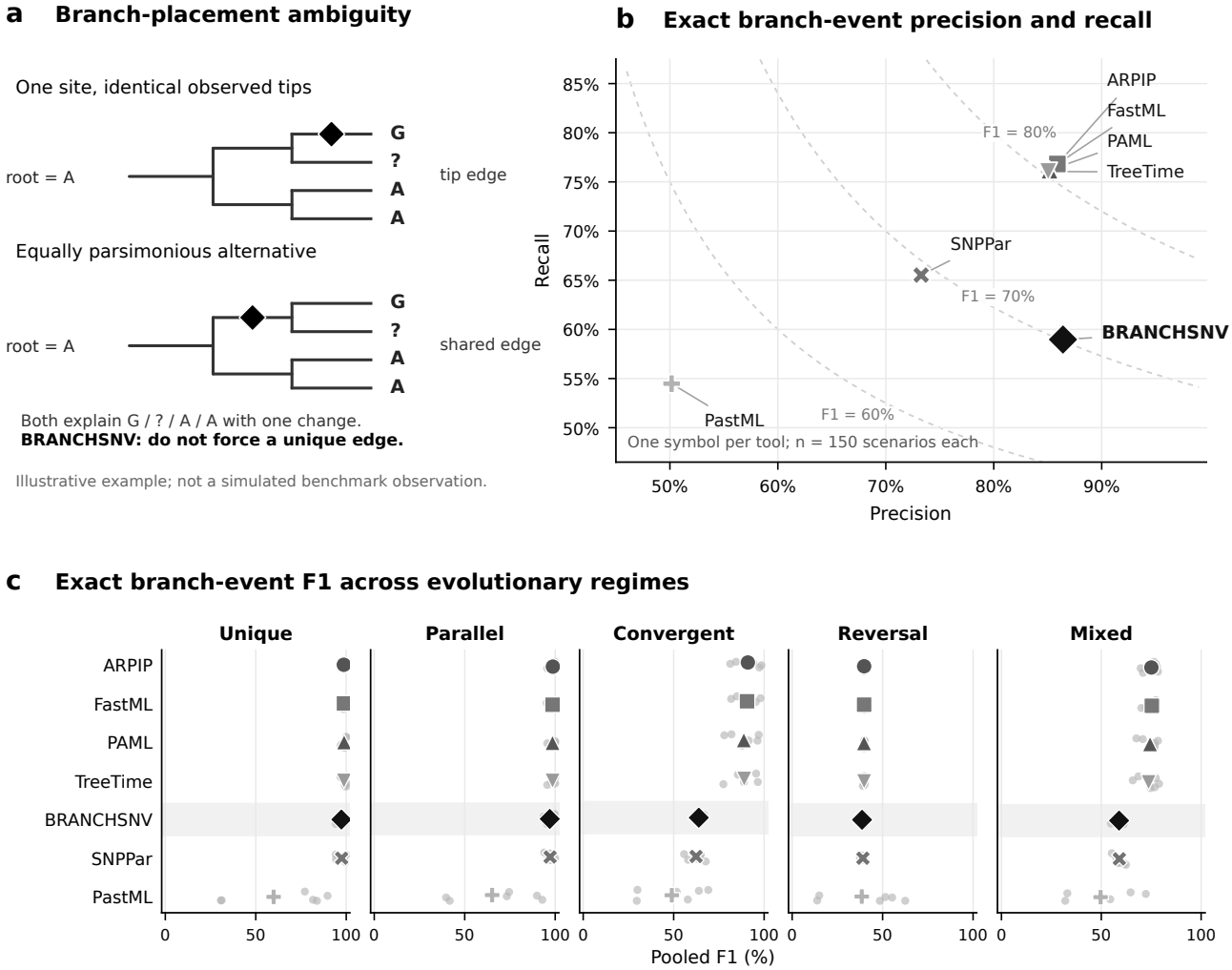
a
Branch-placement ambiguity
One site, identical observed tips
root = A
G
?
A
A
tip edge
Equally parsimonious alternative
root = A
G
?
A
A
shared edge
Both explain G / ? / A / A with one change.
BRANCHSNV: do not force a unique edge.
Illustrative example; not a simulated benchmark observation.
b
Exact branch-event precision and recall
Recall
Precision
85%
80%
75%
70%
65%
60%
55%
50%
50%
60%
70%
80%
90%
ARPIP
FastML
PAML
TreeTime
F1 = 80%
SNPPar
F1 = 70%
BRANCHSNV
PastML
F1 = 60%
One symbol per tool; n = 150 scenarios each
c
Exact branch-event F1 across evolutionary regimes
Unique
Parallel
Convergent
Reversal
Mixed
ARPIP
FastML
PAML
TreeTime
BRANCHSNV
SNPPar
PastML
0
50
100
Pooled F1 (%)
Small grey dots: six five-scenario strata per method and regime; large symbols: pooled F1.
Black diamond = BRANCHSNV; other methods have distinct marker shapes and fixed row labels.